

	33	c
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	34	A
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Question	Marking guidance	Additional comments/guidelines	Mark
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	34	B
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	35	C
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	$P = \Delta mgh/\Delta t$ AND $\Delta m = \rho\Delta V$ seen and combined ✓	condone m for Δm alternative: $P = \text{work done/time} = F\Delta d/\Delta t$ AND $F = (h\rho g)A$ AND $\Delta V = A\Delta d$ seen and combined ✓	1	1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	$140 \text{ (m}^3 \text{ s}^{-1})$ ✓ cao		1	1 × AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Max two: Water still has <u>kinetic energy/KE</u> after passing through turbine ✓ Losses in the generator (eg ohmic heating)✓ Friction in bearings of turbine/generator ✓ reference to fluid friction when passing through the pipes or turbine ✓		2	2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.4	Counts and combines squares (16 below axis, 10 above axis) ✓ Determines the area of one square = 200 MWh OR 7.2×10^{11} J ✓ Net energy = -4.3×10^{12} (J) ✓	For MP1, area below axis must be negative or subtracted POT must be correct for MP2 minus sign required for MP3	3	3 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
03.5	Idea that more energy is required raising water back to the upper reservoir than is generated (over 24 hours) ✓	reject ideas of heat loss in a particular part of the process etc	1	1 × AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
03.6	Stores <u>energy</u> in times of low demand ✓ Quick start up time (during peak demand) ✓		2	2 × AO1
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Use of $I = L/4\pi r^2$ (= 1360 W) ✓ Use of $I = P/A$ (= 400 W) ✓ 0.29 ✓		3	3 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Use of $E(t) = m(t) c \Delta\theta$ ✓ 24 (°C) ✓		2	2 × AO2
Total			5	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Idea that energy is stored in PSS when energy from wind farm exceeds demand ✓ Idea that PSS used when wind farm is unable to meet demand e.g. when wind speeds are low/zero ✓		2	AO2 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Uses the efficiency correctly ✓ Uses $P = \frac{1}{2} \pi r^2 \rho v^3$ ✓ 15 (m s⁻¹) ✓	MP1: e.g. calculates input power $2.3/0.28 = 8.2$ MW. Condone power of ten error for their power. Condone use of diameter for radius 9.3 m s⁻¹ gains 2 marks	3	AO2 AO2 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	$E_p = E_k$ OR $mgh = \frac{1}{2}mv^2$ seen ✓ 110 (m s⁻¹) ✓	No credit for suvat methods.	2	AO2 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.4	momentum of water decreases/changes ✓ (rate of) change of momentum produces a force (from blade on water) ✓ force is exerted on blade by water or correct reference to Newton's 3rd Law ✓ force acts at a distance from centre of mass/axle or relevant reference to $T = Fr$ ✓		4	AO2 AO2 AO3 AO3
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	$2(.2) \times 10^{10}$ (W) ✓	Accept 2×10^{10} to 3×10^{10} (W) Accept 1 sf answer.	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Similarity ✓ Width shows length of time Sun is up and that does not change (much). OR Max of each peak at noon shows brightest time of day. Difference ✓ Different heights show that some days cloudier than others. OR Different shapes show that cloud cover varies from day to day.	If no other mark given, award Max 1 for a similarity and a difference stated. Allow alternatives for 'brightest time of day' e.g. 'that is when the sun is direct', 'at 90°' etc. "Differences in weather" is not enough for the explanation.	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.4	Base-power stations same on both days, AND back-up on day 4 greater (than on day 2) ✓ Base-power: idea that base power stations are on all of the time/solar not available throughout the whole day ✓ Back-up: idea that on day 4 more needed from back-up as less is supplied by solar ✓	Evidence for MP1 may be seen in MP2 and MP3. In MP2 allow idea that base power stations (often) have a long start up time or that their power output is constant.	3	AO3
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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15	A	7 15	AO2
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16	D	3.2%	AO2
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17	C	1.23 W	AO3
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19	D	340 MW	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Divides electrical power by 0.86 ✓ Uses $E = mgh$ or $P(t) = \dot{m}gh$ ✓ 375 or 370 (kg) ✓		3	1 × AO1 1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Uses $P = V_{\text{rms}}I_{\text{rms}}$ or $P = \frac{V_0 I_0}{2}$ or $I_0 = I_{\text{rms}} \sqrt{2}$ ✓ 1.6(2) (A) ✓	If no subscripts seen assume rms. Expect $I_{\text{rms}} = 1.14$ (A) Allow use of both $P = \frac{V^2}{R}$ and $P = I^2 R$ for mp1 If not other marks awarded allow calculation of $V_0 = V_{\text{rms}} \sqrt{2}$	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	Figure 1's station doesn't destroy (large areas of) habitat by flooding or by occupying more area OR Figure 1's station has smaller impact on river because of (temporary) removal of only (a small) part of the flow ✓	Reason must be related to the difference between hydroelectric power stations. Condone the idea that wildlife cannot traverse the dam.	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.4	Base load has to be available all of the time ✓ Max 2 from ✓✓ - Idea that river HEP has variable flow and will not provide a constant output - Idea that river HEP could sustain a constant output if river were reliable enough. - Idea that the reservoir will not sustain constant output if the level drops too much / while water is pumped into reservoir - Idea that reservoir could sustain a constant output if reservoir were big enough / water from dam can be used when (normal) flow is not enough	Ignore start up or shut down times or times to change power levels. Ignore/reject references to amount of power generated. Ignore references to long term and provides base power.	3	1 × AO1 2 × AO2
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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20	A	300 h	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO														
06.1	The mark scheme gives some guidance as to what statements are expected to be seen in a 1- or 2- mark (L1), 3- or 4- mark (L2) and 5- or 6- mark (L3) answer. Guidance provided in section 3.10 of the ‘Mark Scheme Instructions’ document should be used to assist in marking this question.		6	AO1														
	<table><tr><th>Mark</th><th>Criteria</th></tr><tr><td>6</td><td>All three areas covered in some detail. 6 marks can be awarded even if there is an error and/or parts of one aspect missing.</td></tr><tr><td>5</td><td>A fair attempt to analyse all three areas. If there are several errors or missing parts then 5 marks should be awarded.</td></tr><tr><td>4</td><td>Two areas successfully discussed, or one discussed and two others covered partially. Whilst there will be gaps, there should only be an occasional error.</td></tr><tr><td>3</td><td>One area discussed and one discussed partially, or all three covered partially. There are likely to be several errors and omissions in the discussion.</td></tr><tr><td>2</td><td>Only one area discussed or makes a partial attempt at two areas.</td></tr><tr><td>1</td><td>None of the three areas covered without significant error.</td></tr></table>				Mark	Criteria	6	All three areas covered in some detail. 6 marks can be awarded even if there is an error and/or parts of one aspect missing.	5	A fair attempt to analyse all three areas. If there are several errors or missing parts then 5 marks should be awarded.	4	Two areas successfully discussed, or one discussed and two others covered partially. Whilst there will be gaps, there should only be an occasional error.	3	One area discussed and one discussed partially, or all three covered partially. There are likely to be several errors and omissions in the discussion.	2	Only one area discussed or makes a partial attempt at two areas.	1	None of the three areas covered without significant error.
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	Maximum power output • wind speed • radius of blades • ignore references to density of air • importance of location and reliability/strength of wind • reasons not all energy can be used eg friction in bearings, wind speed does not decrease to zero etc																	
Number and arrangement • size of turbine and available space • idea of wind shadows • consideration of size of turbine related to power output and how close together they can be due to wind shadow																		
Environmental factors • noise • harmful to birds • means land can't be used for anything else • aesthetic impact • wind shadow flicker																		

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Determines area of sphere at Earth's orbit OR Multiplies $1.3 (\times 10^3)$ by their area of sphere ✓ 3.8×10^{26} (W) ✓	Condone POT errors in MP1	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Idea that: sunlight reflected from clouds/atmosphere/albedo of Earth is not 0 ✓ angle of Sun changes throughout the day/year ✓ not all power from Sun can be used by solar panel / maximum efficiency is about 30% ✓		3	AO3

Total			11	
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Question	Key	Answer	AO
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16	C	12 V to 18 V	AO4
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